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Differential with disconnect and locking mechanisms

Abstract

A drivetrain component includes a first housing arranged to be rotated by a vehicle motive power source, a second housing, differential gears coupled to the second housing, and first and second clutches that define at least part of a disconnect mechanism and a locking mechanism. The first clutch has a disengaged state in which the first housing rotates relative to the second housing, and the first clutch has an engaged state in which the second housing is coupled to and rotates with the first housing. The second clutch has a disengaged state in which a first shaft can rotate at a different speed than a second output shaft, and the second clutch has an engaged state in which the first shaft and the second shaft rotate at the same speed.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to a differential for a vehicle.

BACKGROUND

(2) Vehicles include differentials that permit wheels of the vehicle to turn at different speeds, such as is needed when the vehicle is turning. Some differentials can include a lock that is engaged to drive wheels coupled to the differential with the same torque, in known manner. And some differentials include a disconnect device by which torque from a vehicle engine can be selectively provided or not to the wheels coupled to the differential so that the vehicle can be operated in 2-wheel drive and 4-wheel or all-wheel drive modes. Significant torque can be routed through some components of the differential in certain operating conditions and this can affect the durability of the differential, or require components of the differential to be larger, heavier and more robust to handle the torque loads.

SUMMARY

(3) In at least some implementations, a drivetrain component includes a first housing arranged to be rotated by a vehicle motive power source, a second housing, differential gears coupled to the second housing, and first and second clutches that define at least part of a disconnect mechanism and a locking mechanism. The differential gears include at least one pinion gear, a first side gear engaged with the pinion gear, and a second side gear engaged with the pinion gear. The first clutch has a first clutch member that is coupled to the second housing, and a second clutch member that is coupled to the first housing. The first clutch has a disengaged state in which the first housing rotates relative to the second housing, and the first clutch has an engaged state in which the second housing is coupled to and rotates with the first housing. The second clutch has a first clutch

member that is coupled to the first side gear and a second clutch member that is coupled to the second housing. The second clutch has a disengaged state in which the first clutch member of the second clutch is not engaged with the second clutch member of the second clutch and the first side gear can rotate at a different speed than the second side gear, and the second clutch has an engaged state in which the first clutch member of the second clutch is engaged with the second clutch member of the second clutch and the first side gear and the second side gear rotate at the same speed.

(4) In at least some implementations, the first clutch member of the first clutch is defined by drive features formed on part of the second housing.

(5) In at least some implementations, the first clutch member of the second clutch is defined by drive features formed on part of the first side gear.

(6) In at least some implementations, the second clutch member of the second clutch rotates with the second housing and is slidably movable relative to the second housing.

(7) In at least some implementations, a first actuator is carried by the first housing and operable to move the second clutch member of the first clutch relative to the first clutch member of the first clutch, and a second actuator is carried by the first housing and operable to move the second clutch member of the second clutch relative to the first clutch member of the second clutch. In at least some implementations, the first actuator is an electromagnetic actuator having a coil that, when powered, produces a magnetic field to drive a drive member that in turn drives the second clutch member of the first clutch relative to the first clutch member of the first clutch. In at least some implementations, the second actuator is an electromagnetic actuator having a second coil that, when powered, produces a magnetic field to drive a second drive member that in turn drives the second clutch member of the second clutch relative to the first clutch member of the second clutch.

(8) In at least some implementations, the differential gears are received within an interior of the first housing, the first housing includes a first shaft mount having a first bore aligned with a first side gear opening of the first side gear so that a first shaft is receivable in the first shaft mount and the first side gear, the first actuator is mounted to the first shaft mount, the first housing includes a second shaft mount having a second bore aligned with a second side gear opening of the second side gear so that a second shaft is receivable in the second shaft mount and the second side gear, and the second actuator is mounted to the second shaft mount.

(9) In at least some implementations, the first shaft mount opening and the second shaft mount opening are coaxially aligned.

(10) In at least some implementations, a drivetrain component includes an outer housing arranged to be driven by a vehicle motive power source, an inner housing received at least partially in the outer housing and capable of rotation relative to the outer housing, a first clutch with a first clutch member connected to the outer housing and a second clutch member connected to or defined by part of the inner housing, differential gears coupled to the inner housing, the differential gears including at least one pinion gear, a first side gear engaged with the pinion gear, and a second side gear engaged with the pinion gear, a second clutch with a first clutch member of the second clutch being coupled to or defined by part of the first side gear, and with a second clutch member of the second clutch being coupled to the inner housing. The first clutch has a disengaged position in which the first clutch member of the first clutch and the second clutch member of the first clutch are not engaged with each other and the inner housing is not coupled to the outer housing by the first clutch, the first clutch as an engaged position in which the first clutch member of the first clutch and the second clutch member of the first clutch are engaged with each other and the inner housing is coupled to the outer housing and torque from the vehicle motive power source is transferred to the inner housing via the outer housing. The second clutch has a disengaged position in which the first clutch member of the second clutch and the second clutch member of the second clutch are not engaged with each other and the first side gear is not coupled to the inner housing by the second clutch, and the second clutch as an engaged position in which the first clutch member of

the second clutch and the second clutch member of the second clutch are engaged with each other and the first side gear and the second side gear rotate at the same speed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following detailed description of preferred embodiments and best mode will be set forth with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a sectional view of a differential including a disconnect mechanism and a locking mechanism;
- (3) FIG. 2 is a diagrammatic view of the differential of FIG. 1 showing the disconnect mechanism in a disconnected state and the locking mechanism in an unlocked state;
- (4) FIG. 3 is a diagrammatic view of the differential showing the disconnect mechanism in a connected state and the locking mechanism in the unlocked state;
- (5) FIG. 4 is a diagrammatic view of the differential showing the disconnect mechanism in a connected state and the locking mechanism in a locked state; and
- (6) FIG. 5 is a diagrammatic view of a differential showing a modified actuator for the locking mechanism.

DETAILED DESCRIPTION

- (7) Referring in more detail to the drawings, FIGS. 1-4 illustrate a drivetrain device shown as a differential **10** that includes a disconnect mechanism **12** and a locking mechanism **14**. The disconnect mechanism **12** permits torque to be selectively provided through the differential **10**. In this way, a vehicle including the differential **10** may be operated, for example, in a 2-wheel drive mode when the disconnect mechanism **12** is in a disconnected state, and a 4-wheel drive mode when the disconnect mechanism **12** is in a connected state and torque is transmitted through the differential **10**. The locking mechanism **14** permits the differential **10** to be operated in an open mode permitting rotation of two output shafts **16**, **18** at different speeds, and in a locked mode in which the output shafts **16**, **18** rotate at the same speed. The differential **10** may be used in a variety of vehicles, including those powered by different motive power devices or sources, such as a combustion engine, electric motor, or both, to provide selective 4-wheel drive and to permit selective locking of the differential **10**.
- (8) The differential **10** includes a first, or outer housing **20**, two or more pinion gears **22**, and a pair of side gears **24**, **26** that are arranged to be coupled to the rotating shafts **16**, **18** that may drive, for example, wheels of a vehicle. Thus, the side gears **24**, **26** rotate with the shafts **16**, **18** (shown diagrammatically in FIG. 1) about a shaft axis **28** and the pinion gears **22** are rotatable about an axis **30** defined by a pinion gear shaft **32** extending through the pinion gears **22**. The pinion gear shaft **32** is connected to a second, or inner housing **34** that is rotatable relative to the outer housing **20**. The outer housing **20** is coupled to and rotates with an outer, ring gear **36** (shown diagrammatically in FIG. 1) that is driven by the vehicle motive power source (e.g. engine, motor or both).
- (9) The differential **10** includes a first clutch **40** that defines at least part of the disconnect mechanism **12** and, when engaged, couples the inner housing **34** to the outer housing **20** so that the inner housing **34** rotates with the outer housing **20**. The first clutch **40** may be of any desired construction and, in at least some implementations, is an engaging clutch such as a dog clutch. The clutch includes a first clutch member **42** that is not movable in an axial direction (defined by the central axis of the plunger **60** and clutch), and a second clutch member **44** that is movable in the axial direction relative to the first clutch member **42** to selectively mesh with, contact or engage the first clutch member **42**. The first clutch member **42** and second clutch member **44** may be annular, and coaxial with the side gears **24**, **26**. The first clutch member **42** may include or be defined by

drive features, such as teeth or other drive features, formed on the inner housing **34** or a component fixed to the inner housing **34**. The second clutch member **44** is coupled to the outer housing **20** and rotates with the outer housing **20**, is movable relative to the first clutch member **42**, and includes mating drive features, such as teeth, that are meshed with the teeth of the first clutch member **42** in the engaged position or engaged state of the first clutch **40**. The second clutch member **44** is driven from a retracted position (FIGS. **1** and **2**) in which the second clutch member **44** is not in contact with the first clutch member **42**, to an advanced position (FIGS. **3** and **4**) in which the second clutch member **44** is in contact with the first clutch member **42**. In the advanced position, the first clutch **40** is engaged and the disconnect mechanism **12** is in a connected state in which the inner housing **34** is coupled to the outer housing **20** so that the inner housing **34** rotates with the outer housing **20**.

(10) The differential **10** also includes a second clutch **46** that defines at least part of the locking mechanism **14** and, when engaged, couples the inner housing **34** to the side gears **24**, **26**, and thus shafts **16**, **18**, so that the shafts **16**, **18** rotate with the inner housing **34**. The second clutch **46** may be of any desired construction and, in at least some implementations, is an engaging clutch such as a dog clutch. The second clutch **46** includes a first clutch member **48** that is not movable in an axial direction (defined by the central axis of the plunger **60** and second clutch **46**), and a second clutch member **50** that is movable in the axial direction relative to the first clutch member **48** to selectively mesh with, contact or engage the first clutch member **48** and define an engaged position or engaged state of the second clutch **46**. The first clutch member **48** of the second clutch **46** may be defined by drive features, such as teeth or other drive features, formed on the adjacent side gear (shown as the first side gear) or a component fixed to the side gear. The second clutch member **50** is connected to the inner housing **34** so that the second clutch member **50** rotates with the inner housing **34** and so that the second clutch member **50** can slidably move relative to the inner housing **34**.

(11) In at least some implementations, the second clutch member **50** may be keyed or splined to the inner housing **34**, with one or more keys/projections received in complementary slots arranged axially (e.g. parallel to the central axis), to permit axial movement of the second clutch member **50** relative to the inner housing **34** and the side gear. The second clutch member **50** could have multiple circumferentially spaced apart and axially extending legs **51** (one is shown and labeled in FIG. **1**) received in openings **53** in the inner housing **34**. When the second clutch **46** is in the engaged state, the differential **10** is in a locked mode and torque is transmitted to both shafts **16**, **18** with relative rotational movement restricted between the shafts **16**, **18** so that both shafts **16**, **18** rotate at the same speed.

(12) In at least some implementations, the first clutch **40** is driven by a first actuator **52** and the second clutch **46** is driven by a second actuator **54** that is separately actuatable and located separate from the first actuator **52**. In the implementation shown in the drawings, both clutches **40**, **46** are driven by electromagnetic actuators. For ease of description and understanding, the components of both actuators **52**, **54** will be given the same reference numerals, as each may be of the same design (although this is not required).

(13) The clutch actuators **52**, **54** have a solenoid with an annular wire coil **58** and a drive member that may include an armature or plunger **60** that may be received at least partially radially inwardly of and axially overlapped with the coil **58**. In at least some implementations, the plunger **60** is also annular, the plunger **60** and coil **58** are coaxially arranged about the axis **28** and are carried by the outer housing **20**. One shaft **16** extends coaxially through a first bore **62** in a tubular first shaft mount **64** of the outer housing **20** over which the coil **58** and plunger **60** of the first actuator **52** are received. The other shaft **18** extends coaxially through a second bore **66** in a tubular second shaft mount **68** of the outer housing **20** over which the coil **58** and plunger **60** of the second actuator **54** are received. The shafts **16**, **18** are coaxially aligned and extend outwardly from the outer housing **20** in opposite directions (e.g. 180 degrees opposed).

(14) The plunger **60** may be formed from multiple materials including a material that is magnetically responsive to the magnetic field generated by the coil **58**, and at least one other material that may or might not be responsive to the magnetic field. Thus, when the magnetic field is generated by the coil **58**, the plunger **60** may be driven from one position to another (e.g. from the retracted to the advanced position). As used herein, a material is responsive to a magnetic field if a magnetic field of the magnitude generated by a solenoid of the type used in applications such as that described herein, may cause a component formed of or including such material to be displaced.

(15) When electric power is supplied to the coil **58** of an actuator **52, 54**, a magnetic field is generated that displaces the plunger **60** relative to the coil **58** and outer housing **20** from a first or retracted position (FIGS. **1** and **2**) to a second or advanced position (FIGS. **3** and **4**). To facilitate return of the plunger **60** from the second position back to the first position when power is not provided to the coil **58**, as shown in FIG. **1**, a biasing member, such as a spring **70** may act on the plunger **60**, or on a disc **72** or other component engaged with the plunger **60**. In the example shown in FIG. **1**, a disc **72** is connected to the second clutch member **44** and has a portion that is engageable by the plunger **60** so that the plunger **60** can drive the second clutch member **44** relative to the first clutch member **42**. The disc may include an outer diameter or a part that extends radially outwardly a sufficient distance to enable position sensing of the disc, and hence the position of the second clutch member **44** to which it is attached, if desired.

(16) In at least some implementations, the clutches **40, 46** are in their engaged states when their respective plunger **60** is in the second position, and the clutches **40, 46** are in their disengaged states when their respective plunger **60** is in the first position. While in the example shown the plungers **60** are in its second position when power is provided to the coils **58** and the plungers **60** move to the first positions when power is not supplied to the coils **58**, the opposite could be true if desired (e.g. either or both clutches could be moved to the engaged position by the respective biasing member and disengaged by powering the respective coil **58**).

(17) In the example of FIGS. **1-4**, the coils **58** are coupled to the shaft mounts **64, 68** and are located outboard of an interior **74** (labeled in FIG. **2**) of the outer housing **20**, where the gears **22, 24, 26**, inner housing **34** and pinion gear shaft **32** are received within the interior **74**. The coils **58** are received in coil housings **76** that are coupled to respective sleeves **78** that have inner surfaces received over an outer surface of the shaft mounts **64, 68**, and with the plungers **60** received for sliding movement along an outer surface of the sleeves **78**. The sleeves **78** may include a radially outwardly extending stop surface **80** that limits movement of the plungers **60** away from their respective first clutch members. An annular retaining member **82** (e.g. clip) may be fixed to the shaft mounts **64, 68** outboard of the sleeves **78**, to retain the sleeves on the shaft mounts, which also retains the position of the coil **58**.

(18) In the example shown in FIG. **5**, the coil **58** of the second actuator **54** is received in a coil housing **84** that is coupled to the outer housing **20** and not to the shaft mount **64**. This coil **58** is more “integrated” with the outer housing **20** and the plunger **60** may slide along the outer surface of the shaft mount **64** without a sleeve **78** between them. Other than the mounting of the coil **58** housing, the actuators **52, 54** of FIGS. **1** and **5** may operate in the same way, with differences only in the magnetic flux path of the generated magnetic fields. Further, while electromagnetic actuators **52, 54** are shown, the clutches could be actuated by other mechanism(s), such as a different linear actuator, or a motor that moves one part of the clutch, for example with one or more gears or cams.

(19) In the example shown, the second clutch member **44** of the first clutch **40** has portions (e.g. feet **86**, labeled in FIG. **2**) that extend axially through openings **87** in the outer housing **20** so that part of the second clutch member **44** is received in the interior **74** of the outer housing **20** and part is outside of the interior **74**. The second clutch member **50** of the second clutch **46** is shown as being fully received within the interior **74** of the outer housing **20** and to move the second clutch member **50**, as shown in FIG. **1**, a drive member **88** is driven by the plunger **60**, where the drive member **88** may be annular and have axially extending portions (e.g. feet) that extend through

openings in the outer housing **20**. The drive member **88** may instead be defined by multiple separate components (e.g. pins) that extend through openings in the outer housing **20** and are circumferentially spaced apart to provide a more uniform actuating force on the second clutch member **50** of the second clutch **46**.

(20) A first disc **90** or washer may be received between the plunger **60** and the pins or other drive member(s) **88** to reduce wear on the components if relative rotation occurs. The disc **90** may also improve uniform movement of the pins **88** during an actuating stroke of the actuator, and the disc **90** may extend radially beyond the coil **58** housing to enable a position sensor to detect the position of the first disc, which relates to the position of the plunger **60** and hence, the clutch. A second disc **92** may be provided between the drive member(s) and the first clutch member **48** of the second clutch **46**, to reduce wear on the components due to relative rotation.

(21) In FIG. **2**, the differential **10** is shown with both clutches **40**, **46** in their disengaged positions. In the illustrated implementation, in the disengaged position of the clutches **40**, **46**, the coils **58** are not powered, the plungers **60** are in their first positions (i.e. retracted positions) and the second clutch members **44**, **50** are not engaged with the first clutch members **42**, **48**. With the first clutch **40** not engaged, the disconnect mechanism **12** is in the disconnected position, and the shafts **16**, **18** are not actively driven by the vehicle motive power source and the shafts may rotate relative to the second clutch member **44** of the first clutch **40** and the outer housing **20**. With the second clutch **46** not engaged, the locking mechanism **14** is in the unlocked position, and the shafts **16**, **18** may rotate at different speeds.

(22) In FIG. **3**, the first clutch **40** is in its engaged position and the second clutch **46** is in its disengaged position. To move the first clutch **40** to its engaged position, the coil **58** of the first clutch actuator **52** is powered to generate a magnetic field that drives the plunger **60** to its second position (i.e. advanced position) which drives the second clutch member **44** into engagement with the first clutch member **42**. In this position, the disconnect mechanism **12** is in the connected state in which the inner housing **34** is coupled to and rotates with the outer housing **20**, and torque is transmitted to the shafts **16**, **18**. Because the locking mechanism **14** is still in the unlocked position (the coil **58** of the second clutch **46** was not powered), the shafts **16**, **18** may rotate at different speeds in an open differential mode.

(23) In FIG. **4**, the first clutch **40** is in its engaged position so the disconnect mechanism **12** is in the connected state as noted with regard to FIG. **3**. The second clutch **46** is in its engaged position. To move the second clutch **46** to its engaged position, the coil **58** of the second clutch actuator **54** is powered to generate a magnetic field that drives the second plunger **60** to its second position (i.e. advanced position) which drives the second clutch member **50** of the second clutch **46** into engagement with the first clutch member **48** of the second clutch **46**. In this position, the locking mechanism **14** is in the locked position and the shafts **16**, **18** rotate at the same speed in a locked differential mode.

(24) As noted above, both the second clutch member **44** of the first clutch **40** and the first clutch member **48** of the second clutch **46** are coupled to the inner housing **34**. In this way, when the disconnect mechanism **12** is in the connected position, torque is transmitted through the inner housing **34** and the differential gears to the shafts **16**, **18** so that the shafts **16**, **18** are actively driven by the vehicle motive power source. Further, when the locking mechanism **14** is in the locked position, the torque flow is more balanced within the device and the disconnect cam ring does not experience loads as high as it would if the locking mechanism **14** were not connected to the inner housing **34** (e.g. if the first clutch member **48** of the second clutch **46** were not coupled to the inner housing **34**). When the disconnect mechanism **12** is in the connected state and the locking mechanism **14** is in the locked state, torque applied to the outer housing **20** flows to: a) the first shaft through the first clutch member **42** of the first clutch **40**, the inner housing **34** (via the second clutch member **44** of the first clutch **40**), the first clutch member **48** of the second clutch **46** (that is coupled to the inner housing **34**) and then to the second clutch member **50** of the second clutch **46**

and to the corresponding side gear; and b) the second shaft through the first clutch member **42** of the first clutch **40**, the inner housing **34** (via the second clutch member **44** of the first clutch **40**), the pinion gear **22(s)** and the other side gear. A locking torque, for example when the vehicle is turning and the side shafts **16, 18** provide reactive torque on their side gears **24, 26** in opposite directions, is provided on the second clutch **46**, the inner housing **34** and the differential gears, but the first clutch member **42** of the first clutch **40** does not directly bear this locking torque. In this way, the first clutch member of the first clutch **40** is not subjected to loads as high as in other arrangements.

(25) The forms of the invention herein disclosed constitute presently preferred embodiments and many other forms and embodiments are possible. It is not intended herein to mention all the possible equivalent forms or ramifications of the invention. It is understood that the terms used herein are merely descriptive, rather than limiting, and that various changes may be made without departing from the spirit or scope of the invention.

(26) All terms used in the claims are intended to be given their broadest reasonable construction and their ordinary meanings as understood by those skilled in the art unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.

Claims

1. A drivetrain component, comprising: a first housing arranged to be rotated by a vehicle motive power source; a second housing; differential gears coupled to the second housing, the differential gears including at least one pinion gear, a first side gear engaged with the pinion gear, and a second side gear engaged with the pinion gear; a first clutch, the first clutch has a first clutch member that is coupled to the second housing and a second clutch member that is coupled to the first housing, the first clutch has a disengaged state in which the first housing rotates relative to the second housing, and the first clutch has an engaged state in which the second housing is coupled to and rotates with the first housing; a second clutch, the second clutch has a first clutch member that is coupled to the first side gear and a second clutch member that is coupled to the second housing so that the second clutch member of the second clutch rotates at all times with the second housing, the second clutch has a disengaged state in which the first clutch member of the second clutch is not engaged with the second clutch member of the second clutch and the first side gear can rotate at a different speed than the second side gear, and the second clutch has an engaged state in which the first clutch member of the second clutch is engaged with the second clutch member of the second clutch and the first side gear and the second side gear rotate at the same speed.
2. The device of claim 1 wherein the first clutch member of the first clutch is defined by drive features formed on part of the second housing.
3. The device of claim 1 wherein the first clutch member of the second clutch is defined by drive features formed on part of the first side gear.
4. The device of claim 1 which also includes a first actuator carried by the first housing and operable to move the second clutch member of the first clutch relative to the first clutch member of the first clutch, and a second actuator carried by the first housing and operable to move the second clutch member of the second clutch relative to the first clutch member of the second clutch.
5. The device of claim 4 wherein the first actuator is an electromagnetic actuator having a coil that, when powered, produces a magnetic field to drive a drive member that in turn drives the second clutch member of the first clutch relative to the first clutch member of the first clutch.
6. The device of claim 5 wherein the second actuator is an electromagnetic actuator having a second coil that, when powered, produces a magnetic field to drive a second drive member that in turn drives the second clutch member of the second clutch relative to the first clutch member of the second clutch.

7. The device of claim 4 wherein the differential gears are received within an interior of the first housing, the first housing includes a first shaft mount having a first bore aligned with a first side gear opening of the first side gear so that a first shaft is receivable in the first shaft mount and the first side gear, the second actuator is mounted to the first shaft mount, the first housing includes a second shaft mount having a second bore aligned with a second side gear opening of the second side gear so that a second shaft is receivable in the second shaft mount and the second side gear, and the first actuator is mounted to the second shaft mount.

8. The device of claim 6 wherein the differential gears are received within an interior of the first housing, the first housing includes a first shaft mount having a first bore aligned with a first side gear opening of the first side gear so that a first shaft is receivable in the first shaft mount and the first side gear, the second actuator is mounted to the first shaft mount, the first housing includes a second shaft mount having a second bore aligned with a second side gear opening of the second side gear so that a second shaft is receivable in the second shaft mount and the second side gear, and the first actuator is mounted to the second shaft mount.

9. The device of claim 7 wherein the first shaft mount opening and the second shaft mount opening are coaxially aligned.

10. A drivetrain component, comprising: a first housing arranged to be rotated by a vehicle motive power source; a second housing; differential gears coupled to the second housing, the differential gears including at least one pinion gear, a first side gear engaged with the pinion gear, and a second side gear engaged with the pinion gear; a first clutch, the first clutch has a first clutch member that is coupled to the second housing and a second clutch member that is coupled to the first housing, the first clutch has a disengaged state in which the first housing rotates relative to the second housing, and the first clutch has an engaged state in which the second housing is coupled to and rotates with the first housing; a second clutch, the second clutch has a first clutch member that is coupled to the first side gear and a second clutch member that is coupled to the second housing, the second clutch has a disengaged state in which the first clutch member of the second clutch is not engaged with the second clutch member of the second clutch and the first side gear can rotate at a different speed than the second side gear, and the second clutch has an engaged state in which the first clutch member of the second clutch is engaged with the second clutch member of the second clutch and the first side gear and the second side gear rotate at the same speed, wherein the second clutch member of the second clutch rotates with the second housing and is slidably movable relative to the second housing.

11. A drivetrain component, comprising: an outer housing arranged to be driven by a vehicle motive power source; an inner housing received at least partially in the outer housing and capable of rotation relative to the outer housing; a first clutch with a first clutch member connected to the inner housing and a second clutch member connected to or defined by part of the outer housing; differential gears coupled to the inner housing, the differential gears including at least one pinion gear, a first side gear engaged with the pinion gear, and a second side gear engaged with the pinion gear; a second clutch with a first clutch member of the second clutch being coupled to or defined by part of the first side gear, and with a second clutch member of the second clutch being coupled to the inner housing, wherein the first clutch has a disengaged position in which the first clutch member of the first clutch and the second clutch member of the first clutch are not engaged with each other and the inner housing is not coupled to the outer housing by the first clutch, the first clutch as an engaged position in which the first clutch member of the first clutch and the second clutch member of the first clutch are engaged with each other and the inner housing is coupled to the outer housing and torque from the vehicle motive power source is transferred to the inner housing via the outer housing, and wherein the second clutch has a disengaged position in which the first clutch member of the second clutch and the second clutch member of the second clutch are not engaged with each other and the first side gear is not coupled to the inner housing by the second clutch, and the second clutch as an engaged position in which the first clutch member of the

second clutch and the second clutch member of the second clutch are engaged with each other and the first side gear and the second side gear rotate at the same speed, and wherein the second clutch member of the second clutch is connected to the inner housing when the first clutch is in either the engaged state or the disengaged state.

12. The device of claim 11 wherein the first clutch member of the first clutch is defined by drive features formed on part of the inner housing.

13. The device of claim 11 wherein the first clutch member of the second clutch is defined by drive features formed on part of the first side gear.

14. The device of claim 11 wherein the second clutch member of the second clutch rotates with the inner housing and is slidably movable relative to the inner housing.

15. The device of claim 11 which also includes a first actuator carried by the outer housing and operable to move the second clutch member of the first clutch relative to the first clutch member of the first clutch, and a second actuator carried by the outer housing and operable to move the second clutch member of the second clutch relative to the first clutch member of the second clutch.

16. The device of claim 15 wherein the first actuator is an electromagnetic actuator having a coil that, when powered, produces a magnetic field to drive a drive member that in turn drives the second clutch member of the first clutch relative to the first clutch member of the first clutch.

17. The device of claim 16 wherein the second actuator is an electromagnetic actuator having a second coil that, when powered, produces a magnetic field to drive a second drive member that in turn drives the second clutch member of the second clutch relative to the first clutch member of the second clutch.

18. The device of claim 15 wherein the differential gears are received within an interior of the outer housing, the outer housing includes a first shaft mount having a first bore aligned with a first side gear opening of the first side gear so that a first shaft is receivable in the first shaft mount and the first side gear, the second actuator is mounted to the first shaft mount, the outer housing includes a second shaft mount having a second bore aligned with a second side gear opening of the second side gear so that a second shaft is receivable in the second shaft mount and the second side gear, and the first actuator is mounted to the second shaft mount.
